GAMSA · Gender-Affirming Medicines South Australia

Ventrogluteal intramuscular administration of testosterone undecanoate 1000 mg/4 mL

Standard operating procedure and training guide

DOCUMENT ID	VERSION	EFFECTIVE	REVISED	NEXT REVIEW
GAMSA-TU-SOP	2.5	5 October 2025	6 November 2026	24 October 2027, or on any change to the Product Information

Current version

Prepared by Theodore South, BPharm (Hons), registered pharmacist, Adelaide. Questions or corrections: gamsaenquiries@gmail.com. A template for adaptation to local policy, offered freely; not endorsed by any health service. Clinics remain responsible for their own procedures, credentialling and emergency arrangements. Where this document and the Product Information differ, the Product Information prevails [1].

At a glance

Site: the gluteal muscle, the only site for which the manufacturer has clinical experience [1][3]; this procedure uses the ventrogluteal site (gluteus medius), SA Health's preferred site for long-acting injections [4].

Needle: draw up with 18G; inject with 21G × 38 mm, or 21G × 50 mm where BMI is over 30 (20–22G acceptable) [5][10].

Technique: 90° to the skin, aspirate, inject the full 4 mL very slowly over approximately 2 minutes, no massage afterwards [1][5][10].

Observe: during and immediately after the injection, and for at least 30 minutes under this procedure, because POME typically presents within 30 minutes [12]; the Product Information gives no fixed time [1][3].

Emergencies: POME — stop, sit up, oxygen, monitor; call 000 immediately for severe breathing difficulty, collapse or signs of anaphylaxis. Anaphylaxis — lie flat, adrenaline 1:1000 IM 0.01 mg/kg up to 0.5 mg, 000 [7][8].

Screen before each injection: current chest infection or breathlessness, previous POME or reaction to an oily injection, bleeding disorder or anticoagulant use (deep IM injection not advisable; seek haematology advice or use a non-injectable product) [1][11]. A pregnancy test before each injection is recommended for anyone who could become pregnant [1][12].

Interval: maintenance every 10 to 14 weeks; the first-to-second interval may be shortened to a minimum of 6 weeks [1][3].

Package contents: this SOP and training guide · treatment-room card (one page) · competency checklist · adverse event form · clinic audit tool. All at https://gamsa.au/medicines/testosterone-undecanoate.

Section 1: Standard operating procedure

Purpose and scope

To ensure the safe, consistent administration of testosterone undecanoate 1000 mg/4 mL (Reandron 1000 and equivalent brands) by deep intramuscular injection into the gluteal muscle, with the ventrogluteal site preferred, minimising the risk of injury and of pulmonary oil microembolism (POME) [1][4][5][6]. Applies to credentialled clinicians (medical practitioners, nurse practitioners and registered nurses) who administer this product in the facility. It also applies to registered pharmacists who have completed accredited training in giving medicines by injection (for example the PSA course), where their state or territory allows it and their pharmacy has the governance in place. In South Australia this is allowed for a medicine that has been prescribed and supplied for the patient. It does not replace the Product Information [1].

Responsibilities

- **Prescribing clinician:** confirms the indication, the dose schedule and informed consent [1].
- **Administering clinician:** trained and assessed as competent in ventrogluteal intramuscular technique and in recognising and managing POME and anaphylaxis (Section 2); competency recorded on the staff file [5].

- **Clinic manager:** ensures emergency equipment (oxygen, adrenaline, monitoring, a means of calling 000) is available and in date, and keeps the current Product Information on file [1][8].

Pre-procedure checks

- Confirm the patient's identity with at least three approved identifiers (for example full name, date of birth, and address or medical record number), and match them to the prescription and the clinical record [13]. Do not use sex or gender as an identifier, because a trans or gender diverse person's records may not all match. Use the name the person uses when you speak to them, and explain why if the legal or Medicare name is needed for a record [12]. Then verify the indication, allergies and informed consent, including any previous reaction to an oil-based injection [1].
- **Screen:** current chest infection or breathlessness (defer if unwell, so that POME is not masked or mistaken); any previous POME or reaction to an oily injection; any bleeding or coagulation disorder or anticoagulant use. The Product Information advises against deep intramuscular injection in those with a bleeding or coagulation disorder, including anticoagulant users, because of the risk of haematoma: use a non-injectable testosterone product or seek a haematologist's advice [1][11].
- **Pregnancy test (recommended):** before each injection for anyone who could become pregnant; testosterone is contraindicated in pregnancy (Product Information sections 4.3 and 4.6) [1][12].
- Confirm the patient can remain for the observation period and has a way home.
- Confirm the product, dose (1000 mg in 4 mL), expiry and batch number. Check the Product Information for contraindications and precautions [1][2].

Equipment

- testosterone undecanoate 1000 mg/4 mL (Reandron 1000 and equivalent brands), one 4 mL vial or ampoule [1].
- 18G needle for drawing up: the solution is a viscous oil [5][10].
- Injection needle 20–22G [10]; this procedure specifies 21G × 38 mm, or 21G × 50 mm where the BMI is over 30, so that the needle reaches muscle [5].
- Alcohol-based hand rub (or a hand basin with soap), 5 mL syringe, gloves, antiseptic swabs, gauze, adhesive dressing, sharps container [5][13].
- Emergency kit: oxygen, adrenaline for intramuscular use, monitoring equipment, a means of calling 000; ASCIA flowchart displayed [8].

Procedure

Clean your hands with alcohol-based hand rub, or soap and water, at each of the 5 Moments for Hand Hygiene: before touching the patient, before the procedure, after the procedure or any body fluid exposure risk, after touching the patient, and after touching the patient's surroundings [14].

1	**Position the patient** side-lying with the upper hip exposed and relaxed, or supine with slight hip flexion. Ensure comfort and privacy [5].
2	**Landmark the ventrogluteal site** with the hand opposite to the hip (left hand for the right hip): heel of the hand over the greater trochanter, index finger towards the anterior superior iliac spine, middle finger spread back along the iliac crest. The centre of the V between the two fingers, over gluteus medius, is the injection point (Appendix A) [4][5].
3	**Clean your hands and prepare the site.** Clean your hands [14], prepare the site with antiseptic and allow it to dry, then put on gloves [5].
4	**Draw up** the full 4 mL with the 18G needle, change to the injection needle and expel air [1][5][10].
5	**Insert the needle at 90°** to the skin, to a depth that places the tip in muscle, not subcutaneous tissue. Aspirate: if blood appears, withdraw, replace the needle and choose a new site [5][10]. Z-track (pulling the skin to one side before the needle goes in) may be used to reduce leakage. It is not part of the Product Information, so follow local policy.
6	**Inject very slowly** over approximately 2 minutes [10]. The Product Information requires very slow injection without giving a time [1][3]; a UK manufacturer's administration guidance for the same formulation specifies about 2 minutes [10]. Slow injection reduces the pressure that forces oil into vessels [6][7].
7	**Withdraw the needle** and apply gentle pressure with gauze. Do not massage the site. Apply a dressing and dispose of sharps immediately [5].
8	**Observe** during and immediately after the injection [1][3], and for at least 30 minutes under this procedure: POME typically presents within 30 minutes [12], and the Product Information gives no fixed time. Book the next appointment before the patient leaves.

Observation and management of reactions

Pulmonary oil microembolism presents within minutes as cough or an urge to cough, breathlessness, malaise, sweating, chest pain, dizziness, paraesthesia or syncope [1][3]. It is uncommon: 28 cases classified as POME in 90,092 doses of the US 750 mg/3 mL product distributed over 4.3 years (2014–2018), a reported rate below 0.1% per injection, against 0.8–2.1% in clinical trials where every injection was observed [6]. Symptoms usually resolve within 30 minutes to a few hours, with supportive care if needed [6][12]. Anaphylaxis is rare but possible [1][3].

Suspected POME

Stop the injection if still in progress. Reassure and sit the patient up (legs outstretched) if breathless [7][8].

High-flow oxygen; monitor pulse, blood pressure, oxygen saturation and respiratory rate [7].

Call 000 immediately for severe breathing difficulty, collapse or signs of anaphylaxis [7][8]. Mild POME symptoms usually settle within 30 minutes to a few hours with supportive care [6][12].

Document, notify the prescriber, complete the adverse event form (Appendix C) and report to the TGA [9].

Suspected anaphylaxis (ASCIA, October 2026)

Lay the patient flat with legs slightly raised; do not let them stand or walk. If breathing is the main problem they may sit with legs outstretched [8].

Adrenaline 1:1000 intramuscularly into the outer mid-thigh: 0.01 mg/kg, up to 0.5 mg (0.5 mL) per dose for an adult. Call 000. Repeat after 5 minutes if there is no improvement [8].

Oxygen, monitoring, and observation in hospital for at least 4 hours after the last dose of adrenaline [8].

Document, notify the prescriber, complete the adverse event form and report to the TGA [9].

Documentation

Record the date and time, product, batch number and expiry, dose, side injected (for example “right ventrogluteal”), needle gauge and length, the administering clinician, any reaction, the patient's condition at the end of the observation period, and the next due date. Maintenance injections are usually given every 10 to 14 weeks; the interval between the first and second injection may be shortened to a minimum of 6 weeks [1][3].

If you have a needlestick or sharps injury, or blood splashes onto you

1. Wash needlestick wounds and cuts with soap and water straight away [16].
2. Rinse splashes to the nose, mouth or skin with water. Rinse eyes with clean water or saline [16].
3. Tell your manager and record it in the clinic's incident system [15].
4. Get medical advice straight away. The doctor decides whether you need treatment to prevent hepatitis B or HIV infection. If hepatitis B immunoglobulin is needed, it must be given within 72 hours [15].
5. Write down the patient's details for the doctor who assesses you. That doctor follows the local procedure for any blood tests [15].

To prevent injury, put each sharp straight into a rigid sharps container at the point of use [15].

Reporting, escalation and review

Any serious or unexpected reaction, especially POME or anaphylaxis, is escalated immediately to the on-call medical officer, recorded in the local incident system and reported to the TGA through AEMS [9]. The event is reviewed, and this procedure or the training updated if needed. The procedure is reviewed annually, or whenever the Product Information, national guidance or local policy changes.

Section 2: Training guide

Learning objectives

On completion the clinician can:

- explain why the gluteal muscle, and the ventrogluteal site in particular, is used for this large-volume oily injection [1][4][5];
- landmark the ventrogluteal site accurately [4][5];
- choose needle gauge and length for the patient's build [5][10];
- inject 4 mL of oily solution deep into muscle over approximately 2 minutes [1][3][10];
- recognise and manage POME and anaphylaxis [6][7][8];
- document and report correctly [9].

Rationale

Reandron 1000 is licensed for deep intramuscular injection into the gluteal muscle, the only site for which clinical experience has been obtained [1][3]. SA Health has moved long-acting injections to the ventrogluteal site because it is safer and absorbs medicine well [4]: it has substantial muscle bulk and lies away from the sciatic nerve and the superior gluteal vessels, which the dorsogluteal site risks [4][5]. The ventrogluteal site accepts the 4 mL volume [5].

Anatomy and landmarking

The site overlies gluteus medius and minimus, bounded by the greater trochanter (lateral), the anterior superior iliac spine (anterior) and the iliac crest (posterior). Landmark as in Section 1 and Appendix A [4][5]. Where landmarks are hard to feel, take time, ask a colleague to confirm, and choose the longer needle so the injection reaches muscle [5].

Needle and equipment selection

Draw up with an 18G needle, as the oil is viscous. Inject with a 20–22G needle [10]; this procedure specifies 21G, 38 mm for most adults and 50 mm where the BMI is over 30 [5]. A wider lumen lets the oil flow with less pressure.

Injection technique

Follow Section 1. The points most often done poorly are injecting too fast (the target is about 2 minutes for the 4 mL [10]), injecting too shallow, and massaging the site afterwards; each increases the chance of oil entering the circulation or leaking back along the needle track [5][6][7].

Recognising and managing complications

- **POME:** sudden cough, urge to cough, breathlessness, chest tightness, sweating, dizziness, paraesthesia or syncope within minutes of injection. Manage as in Section 1. Most episodes are self-limiting; escalate if severe [1][6][7].

- **Anaphylaxis:** manage per ASCIA: lie flat, adrenaline 1:1000 IM outer mid-thigh 0.01 mg/kg up to 0.5 mg (adult), call 000, repeat after 5 minutes if needed, observe at least 4 hours after the last dose [8].
- **Local:** pain, haematoma, oil nodule, infection; manage per local wound care and escalate if infection is suspected [5].

Practical training and competency

- A short teaching session covering the Product Information, POME and anaphylaxis (30–45 minutes) [1][6][7][8].
- Demonstration of landmarking and injection on a model (15–20 minutes) [4][5].
- At least three supervised ventrogluteal injections, documented [5].
- Direct observation against the competency checklist (Appendix B), and a short quiz on recognising POME and anaphylaxis.

Patient information and counselling

- Explain the schedule: usually every 10 to 14 weeks once established [1][3].
- Explain that the injection goes into the side of the hip, takes about two minutes to give [10], and is followed by at least 30 minutes of observation under this procedure (Section 1, step 8).
- Ask the patient to report sudden cough, breathlessness, dizziness, faintness or chest symptoms immediately, during the visit or after it [1][3].
- Give the Consumer Medicine Information leaflet [1]; the plain-language appointment guide below and the Easy Read guide at gamsa.au/easy/injections can be printed for patients.

At the appointment, step by step

1
Lie on your side with the upper hip uncovered, or on your back with that hip bent a little.

2
The clinician finds the spot by hand, using the bony points of your hip. No guessing.

3
The needle goes straight in, and the 4 mL is pushed in very slowly, over about 2 minutes.

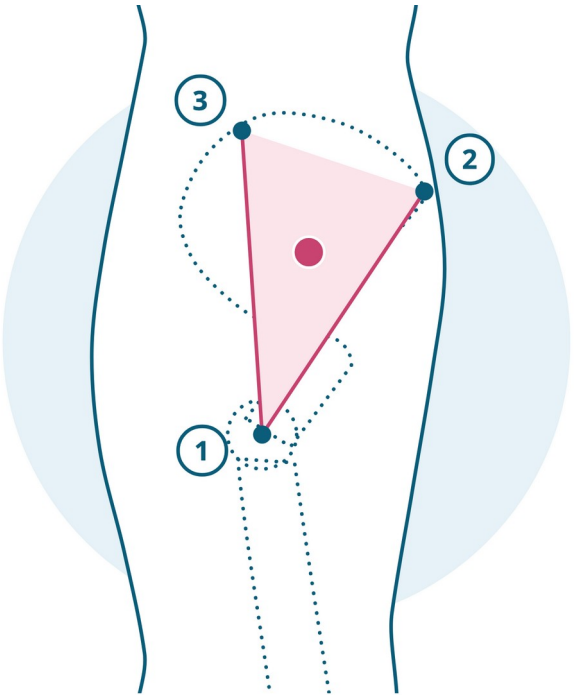
4
Gentle pressure and a dressing afterwards. No hard rubbing or massaging.

5
Important: stay at the clinic for at least 30 minutes after the injection. Tell staff straight away about a sudden cough, breathlessness, chest tightness, dizziness or feeling faint.

6
Book the next one before you leave: about 6 weeks after your first dose, then usually every 10 to 14 weeks.

Figure 1. The appointment, step by step (gamsa.au). Suitable to print for patients.

Appendix A: Landmarking the ventrogluteal site



Where the long-acting injection goes

The ventrogluteal site — the side of the hip

1 Palm flat on the bony knob, fingers pointing up

2 Index finger on the front point of the hip bone

3 Spread your middle finger back along the top ridge

● Needle into the V between them, straight in at 90°

Figure 2. The ventrogluteal site and its three bony landmarks (gamsa.au), drawn from the SA Health brochure [4] and SESLHD guideline [5].

Appendix B: Competency assessment checklist, and Appendix C: Adverse event reporting and escalation form, are separate documents in this package.

References

[1] Aspen Pharmacare Australia. Reandron 1000 (testosterone undecanoate) solution for injection, Australian Product Information, date of revision 11 December 2025. Australian Commission on Safety and Quality in Health Care Medicine Finder. Accessed 18 October 2026. https://www.safetyandquality.gov.au/medicine-finder/reandron-1000

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[9] Therapeutic Goods Administration. Report an adverse event or safety problem; Adverse Event Management System (AEMS) online reporting. Canberra: TGA. Accessed 18 October 2026. https://aems.tga.gov.au/

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[11] ADVANZ PHARMA (Australia). Testosterone ADVZ 1000 (testosterone undecanoate) solution for injection, Australian Product Information, first approved 13 December 2023 (not revised since). Australian Commission on Safety and Quality in Health Care Medicine Finder. Accessed 18 October 2026. (Generic brands Gonadron and Rejunon 1000 are equivalent; check each brand's own PI.) https://www.safetyandquality.gov.au/medicine-finder/testosterone-advz-1000

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Resources

- Patient-facing explanation of the appointment, with the landmarking and appointment infographics (print for patients) https://gamsa.au/using#sec-using-3
- Medicine page: testosterone undecanoate on GAMSA (doses, PBS listing, supply status, this package) https://gamsa.au/medicines/testosterone-undecanoate
- Easy Read guide: Having an injection https://gamsa.au/easy/injections
- ASCIA anaphylaxis action plans and flowcharts (print for the treatment room) https://www.allergy.org.au/hp/anaphylaxis
- Consumer Medicine Information for Reandron 1000 (give to patients) https://www.safetyandquality.gov.au/medicine-finder/reandron-1000-solution-for-injection
- TGA adverse event reporting (AEMS) https://aems.tga.gov.au/

Local adoption record

Adopted or adapted by (service)	Approved by	Date

Amendment history (GAMSA-TU-SOP)

Version	Date	What changed
2.0	2 October 2026	GAMSA edition of the procedure first prepared on 5 October 2025, issued as a package of five documents (the treatment-room card and clinic audit tool were added).
2.1	16 October 2026	Observation after each injection at least 30 minutes (was 15); pregnancy test before each injection recommended for anyone who could become pregnant; POME settling time; reference [12] AusPATH 2026 added.
2.2	18 October 2026	New layout (6 pages); Figure 1 and the landmark key set as text; the POME escalation line is the same in every document; references re-checked (ASCIA October 2026 edition; Product Information links go to the Medicine Finder).
2.3	24 October 2026	Figure 1: step 5 starts with “Important:” and step 6 gives the first interval. Document ID, amendment table, next review date and “uncontrolled when printed” line added.
2.4	31 October 2026	Figure 2 is now the ventrogluteal drawing used on gamsa.au (landmarks numbered 1 to 3, as in the key); Figure 1 step 1 worded as on the site; contact email for questions or corrections added.
2.5	6 November 2026	Three patient identifiers, hand hygiene at the 5 Moments and a needlestick and blood-splash box added; references [13] to [16] added. Z-track noted as optional (follow local policy); registered pharmacists with accredited injection training added to the scope, where their state or territory allows it.